

Algebraic Solution: Oscillator

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2$$
$$= \frac{1}{2} m \omega^2 \left(\hat{x}^2 + \frac{\hat{p}^2}{2m^2 \omega^2} \right)$$

$$\Rightarrow \hat{H} = \frac{1}{2} m \omega^2 V^\dagger V + \frac{1}{2} \hbar \omega \mathbb{1}$$

$$V = \hat{x} + \frac{i\hat{p}}{m\omega}, \quad V^\dagger = \hat{x} - \frac{i\hat{p}}{m\omega}$$

$$\Rightarrow [VV^\dagger] = \frac{2\hbar}{m\omega} \mathbb{1}$$

$$\text{Annihilation: } \hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i\hat{p}}{m\omega} \right)$$

$$\text{Creation: } \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - \frac{i\hat{p}}{m\omega} \right)$$

$$[a a^\dagger] = 1$$

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}^\dagger + \hat{a}) \quad ; \quad \hat{p} = i \sqrt{\frac{m\omega \hbar}{2}} (\hat{a}^\dagger - \hat{a})$$

$$\hat{H} = \hbar \omega (\hat{a}^\dagger \hat{a} + 1)$$

$$\hat{N} = \hat{a}^\dagger \hat{a}$$

$\hat{N}$ eigenstates
are $\hat{H}$ eigenstates

The Annihilation Operator : $\hat{a}$

$$\Rightarrow \hat{a} \psi_0 = 0$$

$$\sqrt{\frac{m\omega}{2\hbar}} \left(x + \frac{\hbar}{m\omega} \frac{d}{dx} \right) \psi_0 = 0$$

$$\hookrightarrow \psi_0(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-\frac{m\omega}{2\hbar} x^2}$$

Operator Manipulation and Spectrum :

Since,

$$\hat{H} = \hbar\omega \left(\hat{N} + \frac{1}{2} \right)$$

$\hat{N}$ eigen states are Energy eigen states.

$$\hat{a} \psi_0 = 0 \Rightarrow \hat{N} \psi_0 = 0$$

$$[\hat{N}, \hat{a}] = -\hat{a}, \quad [\hat{N}, \hat{a}^\dagger] = \hat{a}^\dagger$$

$$\text{Normalised: } \psi_n = \frac{1}{\sqrt{n!}} \cdot \underbrace{\hat{a}^\dagger \dots \hat{a}^\dagger}_{n \text{ times}} \cdot \psi_0$$

$$\hat{a} \psi_n = \sqrt{n} \psi_{n-1}$$

$$\hat{a}^\dagger \psi_n = \sqrt{n+1} \psi_{n+1}$$

&

$$N \psi_n = n \psi_n$$

Q). $\Delta n = ?$ in n -th eigen state

$$(\Delta n)_n^2 = \langle \hat{n}^2 \rangle_{\psi_n} - \langle \hat{n} \rangle_{\psi_n}^2$$

$$\langle \hat{n} \rangle_{\psi_n} = (\psi_n, \hat{n} \psi_n) = \sqrt{\frac{\hbar}{2m\omega}} (\psi_n, (\hat{a} + \hat{a}^\dagger) \psi_n)$$

$$\begin{aligned} a\psi_n &\sim \psi_{n-1} \\ a^\dagger\psi_n &\sim \psi_{n+1} \end{aligned} \left. \vphantom{\begin{aligned} a\psi_n &\sim \psi_{n-1} \\ a^\dagger\psi_n &\sim \psi_{n+1} \end{aligned}} \right\} \text{orthogonal to } \psi_n$$

$= 0$

$(\psi_n, \psi_m) = \delta_{mn}$

↳ or Inner Product. | Norm.

$$\langle \hat{n}^2 \rangle_{\psi_n} = \frac{\hbar}{2m\omega} (\psi_n, (1 + 2\hat{N}) \psi_n)$$

$$= \frac{\hbar}{2m\omega} (1 + 2n)$$